



UNIVERSITY OF ZIMBABWE

FACULTY OF SCIENCE

**DEPARTMENT OF SPACE SCIENCE
AND APPLIED PHYSICS**

**GUIDELINES FOR UNDERGRADUATE
RESEARCH PROJECTS**

TABLE OF CONTENTS

1. Introduction	1
2. Objectives	1
3. Learning Outcomes	1
4. Types of Research Projects	2
5. Organization of the Research Project	2
5.1 Eligibility	2
5.2 Commencement of the Research Project	2
5.3 Duration of the Research Project	2
5.4 Completion of the Research Project	2
5.5 Supervisors	2
5.5.1 Responsibilities of the Supervisor/s	2
5.5.2 Responsibilities of the Student	2
6. Guidelines for the Research Proposal	3
6.1 Introduction	3
6.2 The outline of the research proposal	3
6.3 Description of the Research Proposal components	4
7. Guidelines for the Research Report	7
7.1 Introduction	7
7.2 Requirements for the research project report	7
7.2.1 Binding	7
7.2.2 Length of the report	7
7.2.3 Preliminary pages	7
7.2.3.1 Title page	7
7.2.3.2 Release form	8
7.2.3.3 Approval form	8
7.2.3.4 Dedication (optional)	8
7.2.3.5 Abstract	8
7.2.3.6 Acknowledgements	8
7.2.3.7 Table of contents	8
7.2.3.8 List of tables	8
7.2.3.9 List of figures	9
7.2.4 Chapter 1: Introduction/Background and Justification	9
7.2.4.1 Elements of the introduction (Chapter 1)	9
7.2.4.1.1 Background to the Study	9
7.2.4.1.2 Statement of the problem	9
7.2.4.1.3 Purpose of the study/objectives of the study	10
7.2.4.1.4 Research question (s)	10
7.2.4.1.5 Hypothesis	10
7.2.4.1.6 Significance of the Study	10
7.2.4.1.7 Assumptions of the study	10
7.2.4.1.8 Delimitations of the study	11
7.2.4.1.9 Limitations	11
7.2.4.1.10 Definition of terms	11
7.2.5 Chapter 2: Literature review	11
7.2.5.1 Introduction	11
7.2.5.2 Purpose of literature review	11
7.2.5.3 Key notes about literature review	12
7.2.5.4 Sources of literature review	12
7.2.5.6 Citing references in the text	12
7.2.5.7 General hints for conducting literature review	13
7.2.5.8 Points to emphasize in order to make the literature review	

relevant	13
7.2.6 Chapter 3: Material and Methods/Research Methodology	13
7.2.6.1 Introduction	13
7.2.6.2 Key components of Chapter Three	13
7.2.6.2.1 Introduction	13
7.2.6.2.2 Research design	13
7.2.6.2.3 Subjects	14
7.2.6.2.4 Research instruments	14
7.2.6.2.4 Data collection procedures	14
7.2.6.2.5 Data presentation and analysis procedures	14
7.2.7 Chapter 4: Results and Discussion	14
7.2.7.1 Introduction	14
7.2.7.2 Elements of Chapter 4	14
7.2.7.2.1 Introduction	14
7.2.7.2.2 Data presentation process	15
7.2.7.2.3 Presenting the data	15
7.2.7.2.4 Use of tables and figures	15
7.2.7.2.5 Discussion or interpretation of results	15
7.2.8 Chapter 5: Summary, Conclusions and Recommendations	16
7.2.8.1 Introduction	16
7.2.8.2 Elements of Chapter 5	16
7.2.8.2.1 Summary	16
7.2.8.2.2 Conclusions	16
7.2.8.2.3 Recommendations	16
8. Research Project Assessment	17
8.1 Industrial Research Project	17
8.2 Main Honours Research Project	17
Appendix A: Reference Examples (Havard Style).....	17

1. Introduction

Research is a major component of the academic programmes in the Faculty of Science that are aligned with the government's Education 5.0 philosophy, which emphasises the integration of technology, innovation, and entrepreneurship in the learning process. Education 5.0 strongly implies going beyond the theory, to delivering the actions necessary to achieve the national Vision 2030, which seeks to fundamentally transform Zimbabwe to an upper middle income economy. All faculties within the university are expected to be at the centre of the action, innovatively putting research into use to generate goods and services and crafting practical industrial, business and societal solutions. Research in the Faculty of Science will be carried out through two compulsory Supervised Independent Research modules: Industrial Research Project at Level 3 and the main Honours Research Project at Level 4. Supervised Independent Research has been planned to offer a more in-depth study of a precise topic than what is available in the classroom experience.

Supervised independent research involves intensive, structured projects/dissertations or individual study where a student is required to carry out, under the direction and supervision of a member of academic staff, research, design, innovation, a clinical investigation, artistic or practical work undertaken by a student as a module. It is presented in the form of a project write up, dissertation, report, or in such form that generates desired goods and services as may be designated in the regulations for a specific Programme of study. An academic member of staff regularly interacts with and directs the student's learning outcomes with periodic contact. The assessment of such a module is normally based on continuous assessment, oral presentation and a dissertation or final report submitted to the Departmental Board. Students should be able to learn how to conduct research, use various research methodologies and arrive at significant conclusions. Students are responsible for choosing the subject matter he/she wishes to study, the method, data sources and theoretical question(s), all under the direction of a supervisor.

The Industrial Research Project is typically done during the period of Work Related Learning in Level 3 of study. Students are required to engage in problem solving or innovation research projects in collaboration with Industry and under the guidance of the Academic and Industrial Supervisors. Each student is expected to undertake an Industrial research/Innovation project that solves problems identified during the attachment period or introduces innovations as required by Industry. The Industrial Research Project may be undertaken at the place of attachment, within the faculty or at any of the University's Industrial partner sites, associated research centres or other approved company. Training institutions are encouraged, wherever possible, to offer the student their own in-house research project which may be continued as an Honours project in the final year. The main Honours research project is an in-depth, independent research or creative endeavor, typically undertaken in the final year of all undergraduate (honours) degree programmes. The Honours project demonstrates advanced academic skills like critical thinking, data analysis, and specialized subject knowledge.

2. Objectives

The main objective of the undergraduate research modules is to provide training in research through individual guidance and supervision in the respective fields of study.

3. Learning Outcomes

On completion of the research projects, the student will be able to;

- i. Identify research problems,
- ii. Critically review literature and background information,
- iii. Formulate objectives, design and write research proposals,
- iv. Collect data using appropriate data collection tools,

- v. Analyze and interpret data by using appropriate data analyzing methods,
- vi. Make conclusions based on the findings,
- vii. Compile research reports,
- viii. Defend own research works,
- ix. Disseminate research findings.

4. Types of Research Projects

Undergraduate research projects vary by discipline, but typically include laboratory-based experiments, computational analyses, systematic reviews of existing literature, field-based investigations, or any other types approved by the Department.

5. Organization of the Research Project

5.1 Eligibility

The student should have registered for the appropriate level of study.

5.2 Commencement of the Research Project

Within four weeks of commencement of the first semester.

5.3 Duration of the Research Project

The research project/thesis/dissertation will take two semesters to complete.

5.4 Completion of the Research Project

A student should complete the research project by the eighth (8th) week of the second semester.

5.5 Supervisors

Based on the student's topic of study, the student must choose one member of staff as an academic supervisor/advisor (for Industrial Research Project the student will also need an industrial supervisor/advisor). In some cases the Departmental Board may also appoint a supervisor for each student and if necessary, a co-supervisor may be appointed. The maximum and minimum numbers of students for each supervisor will be decided by the Board every year on the basis of designation. During the period of the research project, the student shall carry out the research under the guidance of the assigned supervisor unless otherwise a change is made due to an unforeseen reason.

5.5.1 Responsibilities of the Supervisor/s

- a) Should provide necessary guidance for the research project
- b) Should maintain regular contact with the student
- c) Should meet the student regularly (e.g. at least once a week) to discuss progress of research work and provide suggestions or feedback timely
- d) Should guide the students to use time productively
- e) Should arrange for alternative supervision during periods of leave or absence
- f) Should confirm and certify the research project report/dissertation
- g) Should guide the student to prepare the presentation

If there is any conflict of interest with the student, he/she should inform the Departmental Board.

5.5.2 Responsibilities of the Student

- a) Should be prepared to convince the supervisor that he/she is highly motivated, capable of working on his/her own, and have a sincere interest in his/her proposed work.
- b) Should submit progress of research work and written work to supervisor/s timely.

- c) Should meet and discuss with the supervisor regularly (e.g., at least 2 hours per week) and maintain a record
- d) Should conduct the research work according to the time work plan.
- e) Should submit the final research report on or before the stipulated deadline.

6. Guidelines for the Research Proposal

6.1 Introduction

- A crucial part of the research process is the research proposal.
- A research will only be as good as the research proposal presented.
- A poor research proposal may mean poor project results. Students are encouraged to have their supervisory committee in place and to start working on their proposals by the middle of the first semester.
- Prior to starting the research, students must go through the proposal stage, during which they will develop their proposal and have it reviewed by their research advisors and the Board of Space Science & Applied Physics.
- Students are not allowed to start on a research project without an approved proposal. During the proposal stage, students should discuss their research interests with faculty members, identify a research topic, conduct preliminary literature review and develop a project proposal.
- Students will be expected to make an oral presentation of their research project proposals before proceeding for research.

6.2 The outline of the research proposal

The research proposal, written using given guidelines (e.g., Arial or Times New Roman font size 12 and 1½ lines spacing), should not exceed 3 pages (between 300 and 500 words), excluding the title page, and must include the following:

- a) Title Page
 - i. Faculty;
 - ii. Department;
 - iii. Programme;
 - iv. Field or Area of Study;
 - v. Proposed Research Title;
 - vi. Student Name & Registration Number;
 - vii. Prospective supervisors; and
 - viii. Date of submission.
- b) Introduction/Background
 - Give a background of the research, stating why it is important to carry out this research.
- c) Problem Statement
 - Show through a review of scientific literature that a gap in knowledge exists that makes your research necessary.
- d) Aims and Objectives
 - Clearly define the project's aims and objectives.
- e) Hypotheses and/or Research Questions (optional)
 - Identify the project's hypothesis(es) and/or research question(s) based on your understanding of the topic/problem
- f) Research Methods
 - Describe and justify the project design, approach and methods
- g) References
 - A limited number of the most important references

6.3 Description of the Research Proposal components

There is no single format for research proposals. This is because every research project is different. However, the suggested outline above includes all that is generally required in a proposal. In this section, the various components (sections) of the research proposal are described further.

a) Title Page

- i. Programme;
- ii. Title of project;
- iii. Name of student (and supervisors); and
- iv. Date (of submission).

b) Introduction/Background

The introduction provides readers with the background information for the research. Its purpose is to establish a framework for the research, so that readers can understand how it is related to other research. In other words, the introduction should lay the broad foundation for the subject area that in turn places the study within the larger context of the scientific literature. There should be evidence of extensive review of literature in the introduction. It should establish the need for the research and indicate that the writer is knowledgeable about the area. The review of literature accomplishes serves several purposes such as.

- i. It shares with the reader the results of previous that are closely related to the study being undertaken;
- ii. It relates a study to the larger, ongoing dialogue in the literature about a topic, filling in gaps and extending prior studies;
- iii. It provides a framework for establishing the importance of the study, as well as a benchmark for comparing the results of the study with other findings; and
- iv. It “frames” the problem earlier identified.

c) Statement of the Problem

The problem statement describes the context for the study within the scientific body of literature. A knowledge gap constitutes the research problem. In other words, a problem is defined as a gap that exists in the literature, theory, or practice that leads to a need for the study. The literature review should then help in the identification of a gap in knowledge which is important to justify the research. It is important in a proposal that the problem stand out—that the reader can easily recognize it. Sometimes, obscure and poorly formulated problems are masked in an extended discussion. In such cases, reviewers and/or committee members will have difficulty recognizing the problem. State the problem in terms intelligible to someone who is generally sophisticated but who may be relatively uninformed in the area of your investigation. Effective problem statements answer the question: Why does this research need to be conducted? If a researcher is unable to answer this question clearly and succinctly, then the statement of the problem will come off as ambiguous and diffuse. In other words, the statement of the problem should address the question: What is it that we do not know about what you are studying that we need to know? It is important to note that a research problem is not a societal problem!. It is a scientific problem which should be studied following the scientific method. If answer to the research problem/question is obvious, it's a management issue that may be resolved without further research.

d) Aims and Objectives

It is important to state the aims and objectives clearly and concisely.

Research Aim

The aim is the broad, overarching purpose of the project. The project aims gives:

- (i) The Broad Purpose:

- It is a general statement of what your research project is intended to accomplish.
- (ii) The Overall Direction:
- It signals the ultimate goal or what you hope to achieve by the end of the project.

Examples:

- To develop an understanding of research issues, to promote the development of research capability, or to give experience in carrying out an independent project

Research Objectives

The objectives are specific, actionable steps that outline exactly how you will achieve that aim. They give:

(i) Specific Steps:

- These are the concrete, measurable, and achievable actions you will take to fulfill your aim.

(ii) Roadmap:

- Objectives provide a detailed plan and a roadmap for how you will collect and analyze data.

Objectives should be measurable, specific and attainable within a specific timeframe. They are Action-Oriented and are often phrased using action verbs.

Examples:

- To review existing literature on the topic.
- To identify specific target groups for investigation.
- To design and test experimental methods.
- To collect and analyze data to answer a defined research question.
- To develop critical thinking, communication, and technical skills through the project.
- To develop three-dimensional pelvis models to analyze the influence of bone material properties on deformation during implant insertion
- To design and optimize novel insulating materials with enhanced thermal properties and reduced environmental impact

e) Hypotheses/Research Questions

A hypothesis is a testable statement and it is central to scientific research. One or more hypotheses could be tested in a study. Note that there are two classes of hypotheses: (i) hypothesis for differences and (ii) hypothesis for relationships. In the case of the former you may intend to test whether there is a difference in the maize yield between two or more areas receiving different rainfall amounts. In the case of the latter, you may intend to test whether there is a relationship between levels of primary productivity and annual rainfall.

Research Questions are most often used in qualitative inquiry, although their use in quantitative inquiry is becoming more prominent. A *research question* poses a relationship between two or more variables but phrases the relationship as a question; a *hypothesis* represents a declarative statement of the relations or differences between two or more variables.

Deciding whether to use questions or hypotheses depends on factors such as the purpose of the study, the nature of the design and methodology, and the audience of the research, etc. The practice of using hypotheses was derived from using the scientific method in science inquiry. They have philosophical advantages in statistical testing.

f) Justification/Purpose of the study

The purpose statement should provide a specific and accurate synopsis of the overall purpose of the study. The purpose statement can also incorporate the *rationale for or justification* of the study. When preparing a purpose statement try to incorporate a sentence that begins with “The purpose of this study is . . .” This will clarify your own mind as to the purpose and it will inform the reader directly and explicitly. The justification of the study is closely linked to the statement of the problem in that the you need to justify the purpose of closing the scientific gap identified during literature review. The justification is in two parts:

- i) Why is the study important in the advancement of science for science’s sake, and
- ii) How will the study benefit society.

In other words you have to answer the question: Why do we need to know that which we do not know? The justification for the study if well-crafted provides the basis funding of the research or lack of it.

g) Materials and Methods

The methods section is really the heart of the research proposal. The activities should be described with as much detail as possible, and the continuity between them should be apparent. Indicate the methodological steps you will take to answer every question or to test every hypothesis illustrated in the Questions/Hypotheses section. In this section, indicate the variables you propose to control and how you propose to control them, experimentally or statistically, as well as the variables you propose to randomize, and the nature of the randomizing unit.

Be aware of possible sources of error to which your design exposes you. You will not produce a perfect, error free design (no one can). However, you should anticipate possible sources of error and attempt to overcome them or take them into account in your analysis. Moreover, you should disclose to the reader the sources you have identified and what efforts you have made to account for them.

h) Timescale/Work plan

Provide a general outline of the time schedule you expect to follow. You need to demonstrate an awareness of the need for planning and the timescale of the research. Inexperienced researchers tend to underestimate the amount of time that the various stages of research will take. Be generous when working out time frames and check them with a more experienced researcher.

i) Source(s) of Funding for Research

Outline the funds and materials that will be needed for your research. Also indicate what funds are available or will be available for your research

j) References

Follow the Faculty of Science guidelines regarding use of references in text and in the reference list. Only references cited in the text are included in the reference list; however, exceptions can be found to this rule. For example, committees may require evidence that you are familiar with a broader spectrum of literature than that immediately relevant to your research. In such instances, the reference list may be called a *bibliography*.

k) Appendices (if any)

The need for complete documentation generally dictates the inclusion of appropriate appendices in proposals (although this is generally not mandatory). The following materials are appropriate for an appendix: verbatim instructions to participants, original scales or questionnaires, permission in writing to reproduce copyrighted instruments from the copyright holder or proof of purchase of the instrument,

interview protocols, sample of informed consent forms, cover letters sent to appropriate stakeholders or official letters of permission to conduct research.

7. Guidelines for the Research Report

7.1 Introduction

The main objectives of this section are to

- a) guide the student on the requirements of the Department with respect to academic writing and presentation;
- b) detail the procedures and practices that one has to follow in writing a research report,
- c) facilitate communication and understanding between the student and supervisors and examiners through the production of a well-organized, coherent and complete manuscript.

In this section we highlight the specifications for the presentation of the research project report in order to achieve uniformity and consistency.

7.2 Requirements for the research project report

The key considerations for a research report include preliminary (front) pages; research report text; text formatting; back pages material and binding.

7.2.1 Binding

The student must submit two bound copies for marking. The student should also submit an electronic copy of the report to the project supervisor.

7.2.2 Length of the report

The length of the report should not exceed 20 pages (8000 words) inclusive of the Appendices.

7.2.3 Preliminary pages

The preliminary pages precede the introduction and should be numbered in lower case Roman numerals, i.e. i, ii, iii, iv, etc while the body of the research is numbered in Arabic numerals. i.e. 1, 2, 3, etc. In the preliminary pages of the research project, you present the following materials: -

- a) Title page;
- b) Approval form;
- c) Release form;
- d) Dedication;
- e) Abstract;
- f) Acknowledgements;
- g) Table of contents;
- h) List of tables; and
- i) List of Figures.

7.2.3.1 Title page

The title page (Appendix 14) must be arranged as follows:-

- i. Name of institution granting the degree;
- ii. Title of the research project: - brief (15 words or less);
- iii. The title of the research project in capital letters (centre aligned);
- iv. Full name starting with first name, then the middle name(s) and lastly the surname; and
- v. The following statement should be written under the name, centre aligned and systematically arranged:

“A RESEARCH PROJECT SUBMITTED IN PARTIAL FULFILMENT OF THE
REQUIREMENTS FOR THE BSc HONOURS DEGREE IN

vi. Month and year of submission at the bottom, centre aligned.

7.2.3.2 Release form

This is a form that grants the University permission to produce copies of the project and reserves the authors' publication rights.

7.2.3.3 Approval form

This serves as an official acknowledgement and acceptance of the research project as satisfactory. It must be signed by the research project supervisor.

7.2.3.4 Dedication (optional)

This serves as a tribute or recognition to a specific individual or group of individuals.

7.2.3.5 Abstract

The abstract should follow the title page and begin on a new page. It should be concise, but comprehensive (between 250 – 300 words). An abstract is used by potential readers to determine at a glance, the contents of the project. The abstract should contain the following:

- i. A clear statement of the problem under investigation and the purpose of the research;
- ii. A description of the research design used, i.e. the design, sample size and sample composition;
- iii. Information on where and how the data was obtained;
- iv. A description of the data analysis technique(s);
- v. A summary of the findings, conclusions and recommendations.

7.2.3.6 Acknowledgements

In this section you acknowledge the person or persons to whom you are indebted for the guidance and assistance in making your study successful. You must present this section on a separate page.

7.2.3.7 Table of contents

The table of contents should be sufficiently informative with page numbers of all sections and chapters indicated. The preliminaries are typed in lower case. For typing the student should ensure that:

- i. Preliminaries page numbers are typed in Roman numerals;
- ii. Chapter numbers page numbers are typed in Arabic numerals;
- iii. Chapter titles and Chapter numbers are typed in UPPER CASE;
- iv. Sub-headings of each chapter are typed in lower case and single spaced;
- v. Page margins are 3.8 cm on the left and 2.5 cm at the right, top and bottom sides;
- vi. Numbering of the pages should be at the bottom centre;
- vii. References and Appendices are presented in Arabic numerals and are typed in upper case;
- viii. Spelling should follow that of the UK. English Dictionary (the "Spell Grammar" checker facility in the MS-Word can be used); and
- ix. Ensure that typescripts are legible and all fonts as well as formats are consistent throughout.

7.2.3.8 List of tables

- i. List of tables should be typed on a separate sheet in case it covers half a page or more;
- ii. Tables should be numbered consecutively using Arabic numbers, e.g. Table 4.1, Table 4.2, etc., depending on the Chapter in which they are found and typed at the top of the table in the same order they are referred to in the text;
- iii. Adequate and self-explanatory headings should be typed above tables;
- iv. Only the first letter of the first word of the heading and of column headings in the table should be capitalized, except where otherwise necessary;

- v. Acknowledgements of the source should be placed below the tables, e.g. Source: Spear and Rice (2006), where applicable; and
- vi. References for tables should be given in full only in the references list at the end of the research project and not as footnotes to the text.

7.2.3.9 List of figures

- i. List of Figures should be typed on separate sheets in case they cover half page or more;
- ii. Captions should be typed below figures, numbering them sequentially in Arabic numbers (e.g. Figure 1.1, Figure 1.2, Figure 1.3, etc.) depending on the Chapter in which they are found. Refer to them in the text in order of appearance; and
- iii. Only the first letter of captions should be capitalized, except where otherwise necessary.

7.2.4 Chapter 1: Introduction/Background and Justification

In this section of the research guidelines, instructions on the key steps on how to write Chapter One of your research project are given. Students should note that Chapter 1 is entitled "Introduction" or "Background and Justification" in order to emphasize its relative function as illustrated below.

7.2.4.1 Elements of the introduction (Chapter 1)

Chapter 1 is broken down into the following subheadings: -

- i. Background to the Study;
- ii. Statement of the problem;
- iii. Purpose of the Study;
- iv. Research question(s);
- v. Statement of the hypothesis
- vi. Significance of the study;
- vii. Assumptions;
- viii. Delimitations of the study;
- ix. Limitations;
- x. Definition of terms; and
- xi. Summary.

7.2.4.1.1 Background to the Study

- i. Provides context to the proposed study. It should describe the situation in which the problem is encountered. The background places the research study into some intelligible context, touching broadly on some of the issues related to it.
- ii. Is generally informed by information, which led the student to the source of the particular research problem. For example, the student might want to touch on the economic, social, geographical, educational or political context of the problem, or the various dimensions in which it manifests itself.
- iii. The student should identify the gap to be filled by the present study. Where the background to the study depends on the literature review, the literature must be cited.

7.2.4.1.2 Statement of the problem

This sub-section contains a brief, but clear statement of the problem to be solved. Research is usually motivated by the need to understand a problem for a possible solution. Problem statements should thus follow from the background, specifying what is to be studied and the problem. It is important to demonstrate that the problem is critical and warrants a study. One could also discuss the potential consequences if the problem is not addressed. Some of the characteristics of a good research problem statement are that it should be:

- i. Researchable, i.e. it should be possible to investigate it empirically through the collection and analysis of (scientific) data;
- ii. Precise, i.e. written in clear, unambiguous language;
- iii. Resolved through research, that is, the problem chosen offers definite sources of information, which when collected, can answer the key questions sufficiently;

- iv. Fitted into the broader context of current theory and relevant research;
- v. Clearly and logically related to its sub-problems/research questions or hypothesis.

7.2.4.1.3 Aim and objectives of the study

It clarifies the aim(s) or objectives of the study, what the study seeks to accomplish. Objectives are a formal expression of the researcher's intentions. Thus, objectives have to be clear and SMART (Specific, Measurable, Attainable, Realistic and Time-bound). Your intentions to explore, explain, infer or replicate what is already known, must be made clear. These terms are further explained as follows:

- i. *To explore* – is just to find out more about an area which few or no people have ventured into. Exploratory studies are done in areas which are little understood and where the relationship among independent and dependent variables is unknown or only a little is known;
- ii. *To describe* – that is to reveal patterns and trends of situations, or events, objects, phenomena or behaviour. By describing them the hope is that they will be understood better, and so answer the question “why”; and
- iii. *To explain* – is to reveal the linkages among the elements' constitution situations, even and phenomena. It also attempts to answer the question “why”.

7.2.4.1.4 Research question (s)

- i. Research questions are developed from the research problem. Good research questions ought to be open to some more/less definitive answers.
- ii. However, while the question must lend itself to some answer, it must not be totally answered by a simple “Yes” or “No.” It must require you to collect and process research evidence as part of the answer.
- iii. Research questions must specify variables. When research questions are addressed individually, they yield responses which can be reconstructed to make up a complete answer to the main research question. The research questions must be precise.

7.2.4.1.5 Hypothesis

Hypotheses are tentative answers, “intelligent” or “probable answers” to the research questions. A researcher who is not confident of statistical testing of hypothesis is advised not to state hypothesis, but research question instead. Hypothesis can be written in null or alternate (directional) form, i.e. H_0 and H_1 . Hypothesis should be simple and concise. It guides the whole dissertation and is often used as a reference point in the subsequent chapters such that the researcher stays within the context of the research.

7.2.4.1.6 Significance of the Study

- i. You should point out how the solution to the problem or the answer to the research question can influence theory and/or practice.
- ii. You must show why it is worth the time, effort and expense involved to carry out the research.
- iii. You explain the likely practical benefits that the study: who would benefit from the study and in what ways these benefits would be felt?.

7.2.4.1.7 Assumptions of the study

- i. Assumptions are statements of what the research believes to be facts, but cannot be verified.
- ii. Remember that assumptions are not the object of the research, but strengthen the basis of your research.
- iii. Like the significance of the study, these assumptions have practical and theoretical implications.
- iv. Without these assumptions, your research cannot be carried out.

- v. For you to be able to carry out the study, you should hold certain facts about the study as given.
- vi. The assumptions that your study makes influence your research findings.

7.2.4.1.8 Delimitations of the study

These refer to the boundaries of the study, which enable you to clearly point out what is included in the study. Delimitations point out what is included in the study such as the populations or sample size, independent and dependent variables among other factors. A description of both conceptual and practical (physical) boundaries is needed. Delimitations answer the following questions:

- i. What are the concerns of this study?;
- ii. What are not its concerns?;
- iii. How far does it go into the treatment of the given issues and where does it stop?; and
- iv. How wide is the field from which it sources its data?.

7.2.4.1.9 Limitations

- i. These are conditions, which are beyond the control of the researcher, but may place restrictions on the study conclusions and their application to other situations.
- ii. It is not enough just to state these limitations (weakness or constraints) of the study without suggesting compensatory factors for ensuring that the research remains valid and reliable.
- iii. Limitations are weaknesses that are inherent in the research, which the researcher is given credit for pointing out and serve to alert the reader/user of the research about what to take note of when interpreting and generalizing the findings and conclusions of the study.
- iv. Limitations should be comprehensive; hence you need to sufficiently clarify them to the reader.

7.2.4.1.10 Definition of terms

- i. You should identify all terms that need to be defined in order to avoid any misinterpretations.
- ii. These definitions help you to establish the frame of reference with which you approach the research problem.
- iii. The independent and dependent variables to be considered should be defined in operational terms, that is, they should either be observable or measurable so that they can be manipulated scientifically.
- iv. These terms should be employed consistently throughout the research project.
- v. Note that dictionary meanings do not serve adequately in defining terms of a research project.

7.2.5 Chapter 2: Literature review

7.2.5.1 Introduction

You must introduce the readers to the Chapter and what it covers.

7.2.5.2 Purpose of literature review

Literature review refers to the work that the researcher read in an effort to understand the topic and research issues. Literature review should show these two types of literature, i.e.

(i) Theoretical literature

Theoretical literature indicates the variables and relationships, which are relevant to the proposed study. Theories have three key functions: (1) to explain observed facts, (2) predict outcomes of relationships and (3) to summarize knowledge. Where theory is not available, the researcher can present arguments or the sequence of events leading to the stated problem.

(ii) Empirical literature

This refers to the previous studies that are similar to the present study. It involves more than just enumeration or a summary of previous studies, but should also show how these studies relate or differ with the current study, thus establishing the knowledge gap. The gap may be in terms of differences in approaches; theoretical start points; unproven theory or theories, inadequate evidence; apparent contradictions; inconclusive evidence or unsatisfactory methods.

Knowledge of related research enables you to define the frontiers of the research fields. For example, Moya (2023), White (2025) and Gee (2021) discovered that about the research problem under investigation. Investigators Samson and River (2025) added to our knowledge. The current research proposes to go beyond the work of Moya and White (2024) in the following manner.....

7.2.5.3 Key notes about literature review

- i. It is important to note that the review of related theory and research enables the student to place the cited literature in perspective. In other words, the student contextualizes others' research findings to the proposed study.
- ii. The review of literature should always prioritize the establishment of a link between the proposed study and the accumulated body of knowledge for the chosen field of study. Please note that studies without link to the existing body of knowledge seldom make significant contributions to the field. Instead, such studies produce isolated bits of information that are of limited usefulness.
- iii. Reviewing related literature helps the student to limit the problem or research questions; define and clarify the concepts of the study.
- iv. Students are advised to never investigate a study where a very similar study was done before unless there is a valid reason for replicating a previous study. The student might want to investigate a different aspect of the research problem. For example, a study might have been carried out to establish the major causes of high failure rate at a certain small-scale company. The student may replicate a similar study by focusing on a large scale company in another location. However, the review of related literature should consult empirical studies that have been done in the area of study.

7.2.5.4 Sources of literature review

- i. Summaries of theses and dissertations;
- ii. Journals;
- iii. Books;
- iv. Primary documents, e.g. circulars and reports;
- v. Electronic sources (online journals, articles etc.);
- vi. Encyclopedia; and
- vii. Conference papers.

7.2.5.6 Citing references in the text

- i. Cite references by author followed by year of publication with a comma, e.g. (Ajusa, 2024). Seen in the modern context, it is generally accepted that there are three ways to improve the standard of living (Conrad, 2022). During the decade particular attention has been paid to the study of perception for example (Sibanda, 2023; Wood, 2019; Abler *et al.*, 2025).
- ii. For two authors cite them both, e.g. (Khumalo and Dhlomo, 2023), for three authors cite them all, e.g. (Mashonjowa, Baloyi and Manjowe, 2025) and for more than three authors cite the first mentioned author followed by *et al.* e.g. (Mlambo *et al.*, 2022).
- iii. Citations in the text should take the following forms.
 - (a)have been reported by Nhau *et al.*, (2025).
 - (b) Tamburai and Gumbo (2023) found that
 - (c) Other researchers (Haka and Marowa, 2021; Apple, 2022) have indicated that.....

(d) Shona and Shava (2024a, 2024b, 2025) found that.....

7.2.5.7 General hints for conducting literature review

- i. Always begin with the most recent publication and work backwards to earlier publications;
- ii. Use primary sources (e.g. circulars and reports) as far as possible;
- iii. Write the bibliographical data of a source on a card (just one source per card);
- iv. First read the summarized sections of a report (abstract) to determine whether it is relevant to your proposed research. Scan through the source to find the relevant sections and quotations;
- v. Important sentences or paragraphs should be quoted rather than summarized indicating their pages clearly; and
- vi. All notes collected have to be read through again before being synthesized and integrated into a well thought out and systematic report. For this reason, you should interact with the study's relevant literature instead of just regurgitating literature.

7.2.5.8 Points to emphasize in order to make the literature review relevant

Your literature review must focus on the following:

- i. Theory from which the research topic is derived;
- ii. Research hypotheses or questions;
- iii. Research problem and sub-problems, and
- iv. Research gap to be filled by the current research.

7.2.6 Chapter 3: Material and Methods/Research Methodology

7.2.6.1 Introduction

In this Chapter, a clear and concise description of how the research was carried out should be provided. Some, call the Chapter "Materials and Methods", while others refer to it as "Research Methodology". This Chapter entails a vivid description of all the activities and procedures undertaken during the course of the research. Decisions on the most suitable methodology to adopt for your proposed study are informed by your literature review. Chapter 3 is generally written in the past tense.

7.2.6.2 Key components of Chapter Three

The research methodology (materials and methods) section includes the following elements: introduction; research design; subjects (study population and sampling); research instruments; data collection procedures and the summary.

7.2.6.2.1 Introduction

This section briefly spells out the main focus of the chapter concerning the research design research instruments; data collection procedures; data presentation and analysis procedure and the chapter summary.

7.2.6.2.2 Research design

The terms 'design' and 'plan' in research refer to a description of the format and theoretical structure under which the study will be carried out, thus the terms are used interchangeably. This also includes the discussion of steps to be taken in order to safeguard the validity or authenticity of the findings. All research designs are formulated in the context of the four key principles of agricultural research namely replication; control; randomization and standardization. The three main concerns of the research plan (design) are:

- i) To identify the design of the proposed research, e.g. descriptive survey, experimental design, correlation research design, historical research design and case study approach;
- ii) To describe the theoretical framework of the design so as to highlight its benefits and limitations as used in the research; and

- iii) To justify the choice of the design regarding the context in which the study will be conducted.

7.2.6.2.3 Subjects

These are research participants you intend to interact with in your endeavour to generate data for the study. You should describe the target population as well as the sample to be used in the study. Also, perform a thorough discussion of the sampling procedures chosen and the justification for their use in the study.

7.2.6.2.4 Research instruments

These are tools used for collecting data (e.g. questionnaires, interviews schedules and observation guides) in the course of finding solutions to the research problem under investigation. Research instruments should be clearly described so as to bring out their strengths and weaknesses thereby as a way of justifying their selection and suitability to the research. It is important to ensure that: (1). The development of research instruments is done according to the research questions and objectives. (2) Measures taken to control the weaknesses identified should be spelt out so as to ensure the validity and reliability of these instruments and data to be collected.

7.2.6.2.4 Data collection procedures

These are steps taken in administering the research instrument(s) and collection of data from the subjects under study. The steps should be a clear; orderly; justified; deliberate and well thought out strategy meant to enhance timely collection of data. Such steps include:

- i. Making appointments with research subjects through telephones, letters, emails, etc;
- ii. Distribution and administration of research instruments by hand, mail or research assistants; and
- iii. Retrieval of instruments.

7.2.6.2.5 Data presentation and analysis procedures

The section provides the overall products used to organize, describe and analyse collected data. The process entails justification of the choice of the presentation and analysis procedures. The findings are:

- i. Presented into logical themes, tables and/or graphs;
- ii. Described or discussed to reveal patterns and meaning; and
- iii. Analysis of variance may be done using statistical packages such as GENSTAT, SPSS, and STATA, MS Excel among others.

7.2.7 Chapter 4: Results and Discussion

7.2.7.1 Introduction

Chapter 4 is referred to as data presentation, analysis and discussion. The process of data presentation entails the following data manipulation activities: scanning, sifting, organizing and summarizing of collected data. Effective data presentation requires the generation of logical themes, tables and figures. It is expected that your discussion and interpretation of findings will remain unbiased and focused on the research objectives, sub-problems, sub-questions or hypotheses.

7.2.7.2 Elements of Chapter 4

7.2.7.2.1 Introduction

In this Chapter, the student presents the findings and outlines data analysis and interpretation. It is only through the analysis of data into information that the meaning of the research can be understood by the intended audience or user. Data analysis embraces both the qualitative and quantitative type. The data are studied from as many angles as possible in order to explore new facts. Interpretation of analyzed data exposes the true meaning of

the data presented in line with the purpose of the study being undertaken. The interpretation deals with what has been achieved, the lessons learnt and recommendations.

7.2.7.2.2 Data presentation process

After data collection, you proceed to the data presentation process where you are expected to:

- i. **Scan and sift data**—Here you read the data to ensure it is complete, accurate, consistent and relevant. You look for trends, which may emerge and assist in the organization of scanned data into meaningful chunks. Answers to each sub-problem or sub-question should contribute to the answering of the main research question.
- ii. **Organizing data**—You should make sense of data by rearranging it into a manageable format. For example, you may compare responses from various respondents and categorize the responses. You need to identify the patterns of the responses to a question or theme through the use of descriptive statistics like the mean, mode and median or other inferential statistics.
- iii. **Summarize the data**—You should utilize different ways of summarizing large amounts of data such use of tables, graphs and statistical summaries.
- iv. **Answering sub-problems or sub-questions**— Here you should focus on each sub-problem or sub-question separately and select data related to it by using a thematic approach i.e. themes derived from research questions.

7.2.7.2.3 Presenting the data

Data presentation is effectively done using tables, figures and text. It should be structured around sub-themes. Independent and dependent variables to be discussed under background of the subjects usually include socio-demographic data such as age, sex, marital status, academic and professional qualifications (if necessary). After presenting data on the background of the subjects, the researcher should move on to data presentation focusing on specific sub-problems or questions of the study. Themes or sub-headings section should emerge from sub-problems or sub-questions. Note that you should avoid using direct questions as sub-headings as much as possible. Hypothesis testing is done using statistical applications such as GENSTAT, SPSS and STATA among others.

7.2.7.2.4 Use of tables and figures

A narrative of the results is usually enhanced by tables and figures. Tables are used to show rows and columns of numerical data. Figures are used to make a graphical or pictorial presentation of data. When using tables and figures you must observe the following:

- i. The title is placed at the top and bottom for tables and figures respectively;
- ii. For uniformity, Arabic numerals should be used to number tables and figures; A table or figure follows as closely as possible the first reference to it in the report;
- iii. After studying each table or figure the you should write a paragraph or two explaining what it contains, drawing the reader's attention to significant findings; and
- iv. Avoid discussing every entry in the table or figure as such style of presentation is boring and defeats the purposes of using tables and figures.

7.2.7.2.5 Discussion or interpretation of results

- i. The discussion of findings should focus on the results.
- ii. The discussion should tie together findings in relation to theory and review of literature.
- iii. If results support or contradict previous research findings on the topic, this should be stated.
- iv. If results differ from the previous finding, an explanation should be attempted as to why this occurred.
- v. If the study was set up to test hypothesis, the discussion section must report the outcome of each hypothesis using appropriate statistical tests.

- vi. Also, ensure that all the results of the sub-problem or sub-question are discussed.
- vii. The writing style should be precise, simple and direct.

[1] Note on reporting of research findings, interpretation and discussions

In terms of language, be tentative (cautious) about the interpretations and conclusions as illustrated by the following statements:

- i. It would appear that most managers in the study were against gender segregation.
- ii. Results seem to suggest that training positively influences worker performance.

[2] Note on reporting of data and statistics

In terms of language, be more definite when reporting data and statistics as shown by these statements:

- i. The mean and standard deviation were 10 and 2.5 respectively.
- ii. The co-efficient of correlation was 0.8.
- iii. Use P-values for statistical significance.
- iv. If results are statistically significant ($P\text{-value} < \alpha$, where $\alpha = 5\%$ (0.05), reject the H_0 (null hypothesis).
- v. Separate the treatment means using the Least significant difference (LSD) at 5%; 1% and 0.1% respectively.

7.2.8 Chapter 5: Summary, Conclusions and Recommendations

7.2.8.1 Introduction

Chapter 5 presents the summary, conclusion and recommendations of the research. After reading through this final chapter, the audience or reader becomes informed of the research problem which was tackled, the research methodology and its limitations, major findings of the study and their implications for practice.

7.2.8.2 Elements of Chapter 5

7.2.8.2.1 Summary

You should summarize the major findings of the research based on the data presented in chapter four.

7.2.8.2.2 Conclusions

These are summed up answers to sub-problems stated in Chapter One. However, conclusions should be drawn from the data reported in Chapter 4, hence they are termed research-based conclusions. For example, the conclusion of the research study on major causes of staff turnover in the hotel industry can be highlighted as follows:

- i. The study found out that although both sexes of workers are involved, males tend to be more affected than females. The reason given is that males tend to be more intolerant to what they view as authoritarian leadership styles than their female counterparts;
- ii. A close association between high staff turnover and poor workers' results on the part of employees was also established. The poor results could be attributed to insufficient training and long working hours; and
- iii. While problem of staff turnover was quite prevalent, it was also observed that nothing was being done by the head office to address this issue.

7.2.8.2.3 Recommendations

Here the researcher examines his or her findings in the light of suggested applications.

8. Research Project Assessment

8.1 Industrial Research Project

ITEM	CONTRIBUTION
Report	60%
Project proposal assessment	10%
Oral presentation of research project report	10%
Industrial Supervisor's assessment of project execution	10%
Academic Supervisor's assessment of project execution	10%
Total	100%

8.2 Main Honours Research Project

ITEM	CONTRIBUTION
Report/Dissertation	60%
Project proposal assessment	10%
Oral presentation of research project	10%
Supervisor's assessment of project execution	20%
Total	100%

Appendix A: Reference examples (Harvard style)

This gives the details you will need to include when referencing various types of source material. **The examples given are taken from Pears and Shields (2025). Pears, R. and Shields, G. (2025) *Cite Them Right: The Essential Referencing Guide*. 13th edn. Bloomsbury Publishing. Bloomsbury Study Skills**

For each example:

- **Reference list** refers to the way it would be cited in your reference list or bibliography when using the 'Cite Them Right' Harvard style.
- **In-text citation** refers to the way that a work would be cited either in the body of the text or in footnotes when using 'Cite Them Right' Harvard style.

Resource	Elements to include:	Examples:
1. Book	1. Authors or Editors 2. Year of publication (in round brackets) 3. Title (in italics) 4. Edition (if applicable) 5. Publisher 6. Series and volume number (if applicable)	Authored book: Reference list: Ashbourn, J. (2014) <i>Biometrics in the new world: the cloud, mobile technology and pervasive identity</i>. 2nd edn. Springer. In-text citation: (Ashbourn, 2014) Edited book: Reference list: Nasta, S. and Stein, M.U. (eds) (2020) <i>The Cambridge history of Black and Asian British writing</i>. Cambridge University Press. In-text citation: (Nasta and Stein, 2020) E-book: Where an e-book looks like a printed book (usually PDFs) with publication information and page numbers - cite it in the same way as a printed book (above). Where specific pagination details are not available use the information you have e.g. %, loc, chapter/page/paragraph. Also add the DOI or web address to the full reference. Reference list: Prior, H. (2020) <i>Away with the penguins</i>. Available at: https://www.amazon.co.uk/Away-Penguins-Hazel-Prior-ebook (Accessed: 20 July 2025). In-text citation: (Prior, 2020, 74%) Reference list: Faulkner, W. (2000) <i>Light in August</i>. Available at: https://www.vlebooks.com/vleweb/product/openreader?id=UniReading&isbn=9781446485521 (Accessed: 10 June 2025). In-text citation: (Faulkner, 2000, ch. 7, p. 105) The information you include in the reference will depend on whether the article is published in a journal or a version found in an institutional repository. It will also vary depending on whether the article has page numbers/article reference number Note that it is no longer necessary to include a place of publication in a book reference (update in the 13th ed. of Cite Them Right, August 2025).
2. Journal Article	1. Authors 2. Year of publication (in	A single author: Reference list: Gulddal, J. (2020) 'That deep underground savage instinct: narratives of sacrifice and retribution in

	round brackets) 3. Article title (in single quotation marks) 4. Journal title (in italics, capitalise the first letter of each word except linking words) 5. Volume number 6. Issue number (if present, in round brackets) 7. Page numbers or article reference number (Include the page numbers of the whole article when writing your full citation, not just the pages you have referred to) 8. DOI or web link for articles without page numbers or an article reference number.	Agatha Christie's Appointment with Death', <i>Textual Practice</i>, 34(11), pp. 1803-1821. In-text citation: (Gulddal, 2020) Two authors - include them both separated by 'and' or &: Reference list: Thomas, D. and Tian, L. (2021) 'Hits from the Bong: the impact of recreational marijuana dispensaries on property values', <i>Regional Science and Urban Economics</i>, 87, article number 103655. In-text citation: (Thomas and Tian, 2021) Three authors - include them all, separate the first two with a comma and use 'and' or & before the third author: Reference list: Adeyeye, S.A.O., Ashaolu, T.J. and Idowu-Adebayo, F. (2022) 'Mycotoxins: food safety, consumer health and Africa's food security', <i>Polycyclic Aromatic Compounds</i>, 42(8), pp. 5779–5795. In-text citation: (Adeyeye, Ashaolu and Idowu-Adebayo, 2022) Four or more authors - include them all in the full reference, but for the in-text citation just state the first author, followed by et al. Reference list: Moise, L., Gutiérrez, A.H., Khan, S., Tan, S., Ardito, M., Martin, W.D. & De Groot, A.S. (2020) 'New immunoinformatics tools for swine: designing epitope-driven vaccines, predicting vaccine efficacy, and making vaccines on demand', <i>Frontiers in Immunology</i>, 11, article number 563362. In-text citation: (Moise et al., 2020) Note that we only require a DOI or web address for journal articles which do not have page numbers or a reference number. There is no need to include the DOI or web address for articles with volume numbers and page numbers or an article reference number even if you accessed them online.
3. Chapter in an edited book	1. Chapter author(s) 2. Year of publication (in round brackets) 3. Chapter title in single quotation marks 4. 'in' followed by book author(s)/editor(s) 5. Book title (in italics) 6. Publisher's name 7. Chapter pagination	Include the page extent of the whole chapter when writing your full citation. Put just the pages you have referred to in the in-text citation. Note that it is no longer necessary to include a place of publication in a chapter reference (update in the 13th ed. of Cite Them Right, August 2025). Example: Reference list: Singh, H., Khurana, L.K. and Singh, R. (2018) 'Pharmaceutical development', in Vohora, D. and Singh, G. (eds) <i>Pharmaceutical medicine and translational clinical research</i>. Academic Press, pp.33-46. In-text citation: (Singh, Khurana and Singh, 2018, p.35)
4. Web page	1. Author (person or company that created the webpage) 2. Year of publication or last update (in round brackets). Scroll to the	You can find many different types of information on the Internet. Check that the item you are referencing is not a journal article, book chapter, or another type of publication which you should be citing in a different way. Examples: Webpage created by a person

	bottom of the page but if there is no date put (no date) 3. Page title (in italics) 4. Available at: URL (Accessed: date)	Reference list: Bologna, C. (2018) What happens to your mind and body when you feel homesick? Available at: https://www.huffingtonpost.com/entry/what-happens-mindbody-homesick_us_5b201ebde4b09d7a3d77eee1 (Accessed: 24 June 2025). In-text citation: (Bologna, 2018) Webpage created by an organisation Reference list: World Health Organization (2025) Sodium reduction. Available at: https://www.who.int/news-room/fact-sheets/detail/sodium-reduction (Accessed: 24 June 2025). In-text citation: (World Health Organization, 2020) Webpage without a date Reference list: (Action on Salt, no date) In-text citation: Action on Salt (no date) UK salt reduction. Available at: https://www.actiononsalt.org.uk/reformulation/uk-salt-reduction-timeline (Accessed: 30 March 2025).
5. Conference Papers	1. Author of paper 2. Year of publication (in round brackets) 3. Title of paper (in single quotation marks) 4. Title of conference proceedings: subtitle (in italics) 5. Location and date of conference 6. Publisher 7. Page references for the paper	Reference list: Jones, L. (2018) 'Polymer blends based on compact disc scrap', Proceedings of the Annual Technical Conference - Society of Plastics Engineers. San Francisco, May 6-9. Society of Plastics Engineers. pp.236-254. In-text citation: (Jones, 2018)
6. Secondary Referencing A secondary reference is used when you are referring to a source which you have not read yourself, but have seen quoted or read about in another source. Where possible, you should always try to read the original of anything you wish to refer to; otherwise you are relying on the author who cited the reference to have interpreted it correctly and not taken it out of context. Use the reference list at the end of the source you are reading to find details of the reference and search for it If you cannot get hold of the original source you will need to do a secondary reference and you should make clear that you are not using the original source. Only include the source you have used in your list of references following the guidance for citing that type of publication.		
Examples: If the author quotes another source: France (2003, quoted in Weingart et al., 2018, p. 24) provides evidence that hospitals use internal reporting procedures to identify... If the author summarises another source In-text citation: According to France (2003, cited in Weingart et al., 2018, p. 24), hospitals use internal reporting procedures to identify... In both examples only the full reference for the article by Weingart et al. would be included in the reference list.		